

August 11 Reduction mod p



We have studied the irreducible factorization
in $\mathbb{Z}[X]$ with the help with $\mathbb{Q}[X]$.

That is, a polynomial of positive degree
in $\mathbb{Z}[X]$ is irreducible iff it is primitive
and it is irreducible in $\mathbb{Q}[X]$.

and we often have this picture

$$\mathbb{Z}[X] \subseteq \mathbb{Q}[X] \subseteq \mathbb{R}[X] \subseteq \mathbb{C}[X] \quad \text{in mind}$$

Now we consider another direction:

descending from $\mathbb{Z}[X]$ to $\mathbb{Z}_p[X]$

for p some prime number in $\mathbb{Z}$.

Note $\mathbb{Z}_p$ is a finite field and hence

finitely many polynomials for each degree.

So technically we can check the irreducibility

of a polynomial in $\mathbb{Z}_p[x]$ by enumerating

all the irreducible factors that has a smaller degree.

Example: Consider $\mathbb{Z}_2[x]$. All degree 2 polynomials

are- x^2 , x^2+1 , x^2+x , x^2+x+1

and note 0 is a root to x^2

and 1 is a root to x^2+1 , x^2+x .

So $x \mid x^2$ and $(x-1) \mid x^2+1$ and x^2+x

The only irreducible deg 2 polynomial is

$$x^2+x+1.$$

Consider x^5+x^2+1 . Note both 0 and 1

are not roots, So x^5+x^2+1 has no

roots in $\mathbb{Z}_2$. That is, no linear polynomial

divide x^5+x^2+1 . Any deg 0 polynomial

in polynomial ring over a field is irreducible

So x^5+x^2+1 is reducible iff

it factors into factors of strictly

smaller degrees. We showed no deg 1

factors divides $x^5 + x^2 + 1$. It remains to

show no irreducible deg 2 factors

Clearly $x^2 + x + 1$ does not divide $x^5 + x^2 + 1$

Then $x^5 + x^2 + 1$ is irreducible in $\mathbb{Z}_2[x]$

Lemma 2: Suppose $f(x) \in \mathbb{Z}[x]$ factors

as $f(x) = g(x)h(x)$, for $g(x), h(x) \in \mathbb{Z}[x]$.

Then $[f](x) = [g](x)[h](x)$.

in $\mathbb{Z}_p[x]$ for p a prime in $\mathbb{Z}$.

Here if $f(x) = \sum_{i=0}^n a_i x^i$ in $\mathbb{Z}[x]$

then $[f](x) = \sum_{i=0}^n [a_i] x^i$ in $\mathbb{Z}_p[x]$

Corollary 3: Let $f(x) \in \mathbb{Z}[x]$ be of positive

degree, and p be a prime in $\mathbb{Z}$ that

does not divide the leading coefficient

of $f(x)$. If $[f](x)$ is irreducible in $\mathbb{Z}_p[x]$.

then $f(x)$ does not factor as a product

of lower degree polynomials.

proof. Suppose $f(x) = g(x)h(x)$ for $g(x), h(x) \in \mathbb{Z}[x]$

with $\deg g(x), \deg h(x) < \deg f(x)$.

In particular, let $a_t = LC(g(x))$

and $b_s = LC(h(x))$ for $t, s < \deg f(x)$.

Then $LC f(x) = a_t b_s$ and by assumption

$p \nmid a_t b_s$. This implies $p \nmid a_t$ and

$p \nmid b_s$. Hence $\deg [g](x) = \deg g(x)$

and $\deg [h](x) = \deg h(x)$.

Now $[f](x) = [g](x)[h](x)$ in $\mathbb{Z}_p[x]$

By assumption, $[f](x)$ is irreducible

in $\mathbb{Z}_p[x]$. So wlog, we assume.

$[g](x)$ is a unit in $\mathbb{Z}_p[x]$ which implies

$\deg [g](x) = 0$. This implies $\deg g(x) = 0$

and $\deg h(x) = \deg f(x)$. a contradiction.

Remark: In the setting of Corollary 3

if $f(x)$ is also primitive, then $f(x)$ is

irreducible in $\mathbb{Z}[x]$.

This process is called reduction mod p , which is closely related to Gauss's lemma and Eisenstein criterion.

Proposition 4: (Gauss's lemma) Let $f(x), g(x)$ be primitive in $\mathbb{Z}[x]$. Then $f(x)g(x)$ is primitive.

proof. Note that $f(x)$ (and $g(x)$ respectively) is primitive iff $\forall$ prime $p \in \mathbb{Z}$,

$\exists$ some coefficient of $f(x)$ that is not divisible by p . iff $[f](x) \neq 0$ in $\mathbb{Z}_p[x]$ for all prime p in $\mathbb{Z}$. So it suffices

to show $f(x)g(x)$ has the property that

$[f](x)[g](x) \neq 0$ in $\mathbb{Z}_p[x] \forall$ prime $p \in \mathbb{Z}$.

which is clear since $\mathbb{Z}_p[x]$ is always a domain.

Proposition 5: (Eisenstein criterion).

Let $f(x) = a_n x^n + \dots + a_0$ be in $\mathbb{Z}[x]$ and

$n > 1$. Suppose $p \in \mathbb{Z}$ is a prime such that.

1. $p \nmid a_n$.

2. $p \mid a_i \quad \forall 0 \leq i < n$

3. $p^2 \nmid a_0$

Then $f(x)$ does not factor as a product of
polynomial of smaller degrees in $\mathbb{Z}[x]$.

In addition, if $f(x)$ is primitive.

then $f(x)$ is irreducible in $\mathbb{Z}[x]$.

proof. Let $f(x)$ be of in the assumption.

and suppose $f(x) = g(x)h(x)$ for $g(x), h(x) \in \mathbb{Z}[x]$

and $\deg g(x), \deg h(x) < n$. then.

consider $[f](x) = [g](x)[h](x)$ in $\mathbb{Z}_p[x]$.

where $[f](x) = x^n$. It follows

$[g](x) = x^s$ and $[h](x) = x^t$ in $\mathbb{Z}_p[x]$

for some $s, t < n$. Indeed, $\mathbb{Z}_p[x]$ is an UFD

and $x^n = x \cdot x \cdots x$ is the unique

irreducible factorization. In particular,

the constant terms of $g(x)$ and $h(x)$ is divisible

by p . which is impossible since

$p^2 \nmid a_0$.